VIT BHOPAL UNIVERSITY

VITyarthi E-Learning Platform

CSE0001 – Digital Literacy

PROJECT REPORT

Digital Literacy Portfolio

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Declaration of Originality

I, Aditi Pandey (Reg.no: 25BCE11306), hereby declare that this project report titled 'Digital Literacy Portfolio' submitted for the course CSE0001 – Digital Literacy at VIT Bhopal University is an original work carried out by me under the guidance of my faculty mentor. The content presented in this report reflects my own understanding, observations, and reflections developed through hands-on engagement with each task.

I further declare that:

- All written reflections are in my own words and have not been copied from any external source.
- Screenshots submitted as evidence are of my own accounts and work, and have not been borrowed from any other student.
- Where external sources, articles, or platforms have been referenced, they have been duly cited in the References section.
- This submission is my individual work. Although I may have discussed concepts with peers for better understanding, the final deliverables are entirely my own.
- I am aware that any violation of academic integrity norms may result in a Fail grade for this project.

Signature: _____

Name: Aditi Pandey

Date: 31 March 2026

Acknowledgement

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I am also grateful to the VITyarthi E-Learning Platform for providing clear instructions, module resources, and supplementary materials that made it possible to complete all five tasks systematically — even without prior technical experience.

My appreciation goes to my classmates and batchmates who provided moral support and engaged in meaningful peer discussions throughout the course. These conversations helped me refine my understanding of concepts such as digital communication, professional online presence, and cybersecurity awareness.

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Executive Summary

This report documents the complete Digital Literacy Portfolio project undertaken as part of the CSE0001 – Digital Literacy course at VIT Bhopal University. The project spans five interconnected tasks, each aligned with a module of the course, and collectively forms a comprehensive exercise in building foundational digital skills for the modern academic and professional world.

Task 1 involved the creation of a one-page Digital Literacy Awareness Infographic using Canva, covering the concepts of digital literacy, safe internet practices, and professional online presence. The exercise introduced principles of visual communication and information design.

Task 2 required setting up a professional Student Digital Portfolio across three platforms: GitHub, LinkedIn, and Kaggle. This task provided hands-on experience in creating and managing a digital identity that can grow alongside an academic and professional journey.

Task 3 focused on Coding and Collaboration Platforms. Part A involved completing a beginner challenge on HackerRank, while Part B required building a five-question Digital Literacy Awareness Quiz using Google Forms, linked to a Google Sheet for response tracking.

Task 4 addressed Professional Email Etiquette, requiring the drafting of two formal emails — one to a professor and one to an internship coordinator — along with a Social Media Do's and Don'ts Checklist covering responsible digital communication.

Task 5 centred on Cybercrime Awareness, involving a case study on UPI/Online Payment Fraud and a comprehensive Stay Safe Online prevention checklist with at least eight actionable tips tailored for college students in India.

All tasks were organised in a structured GitHub repository, and this report serves as the written documentation of the process, learning, and reflections across all five modules. The total marks available for the project are 100, and the pass mark is 50 out of 100.

1. Introduction

1.1 Background and Context

We live in an era defined by digital transformation. From how we communicate and learn, to how we work and collaborate, the digital dimension of life has become inseparable from its physical counterpart. For a first-year engineering student entering higher education in 2025, the ability to navigate digital platforms confidently, communicate professionally online, and protect oneself from digital threats is no longer a supplementary skill — it is a survival necessity.

Digital literacy, as a concept, extends well beyond the ability to use a smartphone or browse the internet. It encompasses a broad set of competencies: understanding how digital tools work, knowing how to evaluate information critically, being able to communicate clearly and ethically in digital spaces, creating content responsibly, and protecting oneself and others from cybercrime. The United Nations Educational, Scientific and Cultural Organization (UNESCO) defines digital literacy as the ability to access, manage, understand, integrate, communicate, evaluate, and create information safely and appropriately through digital technologies for employment, decent jobs, and entrepreneurship.

India's rapidly evolving digital ecosystem makes this literacy even more urgent. With over 900 million internet users as of 2024, India is one of the world's largest and fastest-growing digital markets. The Unified Payments Interface (UPI) alone processes billions of transactions every month. Yet, with this growth comes risk: phishing scams, UPI fraud, identity theft, cyberbullying, and ransomware attacks are increasingly targeting young, digitally active populations — including college students.

It is within this context that VIT Bhopal University has thoughtfully embedded the CSE0001 – Digital Literacy course into the first-year B.Tech curriculum. By requiring all students, regardless of branch, to engage with the digital world through structured tasks and reflection, the course aims to produce graduates who are not merely technically skilled but also digitally responsible citizens.

This project — the Digital Literacy Portfolio — is the culmination of that course. Through the scenario of being appointed as a Student Digital Ambassador, students

are invited to build, document, and reflect upon a set of digital competencies that will serve them throughout their four-year degree and beyond.

1.2 Objectives of the Project

The key objectives of this project are as follows:

- To develop practical digital skills across five domains: visual communication, professional digital identity, coding platforms, email etiquette, and cybercrime awareness.
- To build a structured, publicly accessible GitHub repository as a living digital portfolio.
- To practise professional communication by creating well-crafted email drafts and a social media responsibility checklist.
- To raise awareness of cybercrime risks relevant to Indian college students and develop actionable prevention habits.
- To reflect meaningfully on each task, articulating what was learned, what was challenging, and how the skills will be applied in the future.
- To submit a comprehensive project report that documents the entire learning journey.

1.3 Scope and Limitations

This project covers all five tasks as outlined in the project brief. The scope includes: infographic design, digital portfolio setup across three platforms, completion of a beginner-level coding challenge, creation of a Google Form quiz, drafting of two professional emails, a social media checklist, a cybercrime case study, and a prevention checklist.

The following limitations are acknowledged:

- All platform accounts were created using a college email address. Contact details and photographs were intentionally excluded to maintain privacy.
- The coding challenge completed on HackerRank was at the beginner level, as no prior coding experience was required for this course.
- The cybercrime case study in Task 5 is based on a realistic fictional scenario, as no personal incidents were available or appropriate to share.
- All screenshots are representative; actual account details have been kept minimal and professional in accordance with course guidelines.

1.4 Structure of the Report

This report is organised as follows:

1. Introduction – Background, objectives, scope, and structure.
2. Task 1 – Design and reflection on the Digital Literacy Infographic.

3. Task 2 – Digital Portfolio setup across GitHub, LinkedIn, and Kaggle.
4. Task 3 – Coding practice and Google Forms collaboration.
5. Task 4 – Email drafts, social media checklist, and communication reflection.
6. Task 5 – Cybercrime case study and prevention checklist.
7. GitHub Repository Structure – Overview of the submitted repository.
8. Conclusion – Key takeaways and future plans.
9. References – All sources and platforms cited.

Appendices follow the main body, including a glossary, resource links, and a self-assessment rubric.

2. Task 1 – Digital Literacy Awareness Infographic

Module 1 | 20 Marks | Deliverable: Infographic image in task-1-presentation/ folder + 150–200 word reflection in project report.

2.1 Task Overview and Requirements

Task 1 required me to step into the role of a Student Digital Ambassador and create a one-page visual resource that helps batchmates understand what Digital Literacy means and why it matters. The task was intentionally open-ended in terms of tool choice — students were free to use Canva, Prezi, Miro, Google Slides with a SlidesCarniVal template, or any other free, browser-based design tool.

The specific requirements were:

- Use a free presentation or design tool of choice.
- Create a single infographic or visual slide covering at least 3 of the following topics: what digital literacy is, useful digital tools for students, safe internet practices, professional online presence, and email etiquette.
- Export or screenshot the design and add it to the task-1-presentation/ folder in the GitHub repository.
- Write a 150–200 word reflection describing the tool used, what the design shows, and one thing found difficult or interesting.

This task tested not just creativity but also the ability to simplify and communicate complex ideas visually — itself a core digital literacy skill.

2.2 Tool Selection: Canva

After reviewing the suggested tools, I chose Canva (canva.com) for several reasons. Canva is a cloud-based graphic design platform that requires no installation and is entirely free for basic use. It offers thousands of ready-made templates specifically designed for infographics, presentations, posters, and educational materials.

The decision to use Canva was driven by the following factors:

- Ease of access: Canva works directly in the browser with no software installation required.
- Template library: Canva offers hundreds of professionally designed infographic templates that serve as a starting point, saving time on layout and colour scheme decisions.

- Drag-and-drop interface: Even without prior design experience, Canva's intuitive drag-and-drop system makes it possible to add, move, and resize elements effortlessly.
- Free tier adequacy: The free Canva account provides sufficient features — icons, fonts, backgrounds, and colour palettes — for the scope of this task.
- Export options: Canva allows designs to be exported as PNG, PDF, or JPG files, making it straightforward to add the final image to the GitHub repository.

Canva vs. Other Suggested Tools

To make an informed tool selection, I briefly explored the alternatives before settling on Canva:

- Prezi: Prezi is a presentation tool known for its non-linear, zooming format. While visually engaging for presentations, it is less suited to creating a static one-page infographic. Its learning curve is also steeper for first-time users.
- Piktochart: Piktochart is specifically designed for infographics and is an excellent alternative. However, the free tier of Piktochart limits the number of downloads and watermarks some exports, making Canva the more practical choice.
- Google Slides with SlidesCarniVal: Google Slides is familiar and accessible, but it is fundamentally a presentation tool rather than an infographic creator. While SlidesCarniVal templates improve the visual quality, the results tend to be slide-like rather than infographic-like.

Canva struck the best balance between design quality, ease of use, and export functionality for a beginner-level infographic task.

2.3 Design Process and Methodology

The design process followed a structured approach, beginning with content planning and ending with final export and upload to the GitHub repository.

Step 1: Account Creation and Template Selection

I created a free Canva account using my college email address (22bce1001@vitbhopal.ac.in). After logging in, I searched for 'infographic' in the template library and browsed through the available options. I selected a clean, vertically oriented template with a blue and white colour palette, which I felt conveyed professionalism and clarity — qualities consistent with the message of digital literacy.

Step 2: Content Planning

Before modifying the template, I planned the content for each section. I decided to cover three topics from the list provided in the brief:

10. What Digital Literacy Is – A definition and overview of the concept.
11. Safe Internet Practices – Key habits for staying safe online.
12. Professional Online Presence – Why and how to maintain a professional digital identity.

I chose these three topics because they form a logical progression: first understanding what digital literacy is, then practising it safely, and finally applying it professionally.

Step 3: Visual Design and Layout

The infographic was structured with a bold title at the top — 'Digital Literacy: What It Is & Why It Matters' — followed by three clearly separated sections, each with a distinct background colour block, an icon, a heading, and three to four bullet points.

- Section 1 (blue): 'What is Digital Literacy?' – Defined as the ability to access, evaluate, create, and communicate information using digital technologies.
- Section 2 (teal): 'Safe Internet Practices' – Covered using strong passwords, enabling two-factor authentication, avoiding phishing links, and not sharing personal information publicly.
- Section 3 (navy): 'Professional Online Presence' – Covered keeping social media professional, creating a LinkedIn profile, maintaining a GitHub portfolio, and using a college email for sign-ups.

Icons from Canva's built-in library were used to reinforce each section visually. The font used throughout was Montserrat — a clean, modern sans-serif that is easy to read at small sizes and maintains visual consistency.

Step 4: Review and Export

After completing the design, I reviewed the infographic for spelling errors, alignment issues, and readability. I then exported it as a PNG file at high resolution (300 DPI equivalent) and uploaded it to the task-1-presentation/ folder in my GitHub repository.

2.4 Content Coverage and Visual Strategy

The infographic was designed not just to be visually attractive but to be genuinely useful as a quick reference guide for first-year students. Each section was crafted with the target audience in mind — students who may have heard the term 'digital literacy' but have never had it explained clearly or practically.

Section 1: What is Digital Literacy?

This section opened with a simple, student-friendly definition: 'Digital literacy is the ability to find, evaluate, use, and create information using digital technology.' This definition was deliberately chosen over more academic phrasings because it is memorable and action-oriented.

The section also included a brief note on why digital literacy matters — specifically, that it is increasingly expected by employers, universities, and professional communities, and that it helps students protect themselves online and participate meaningfully in the digital economy.

Section 2: Safe Internet Practices

This section focused on four core safe internet habits relevant to college students:

- Use strong, unique passwords for each account and consider using a password manager.
- Enable two-factor authentication (2FA) wherever possible — especially for email, banking, and social media accounts.
- Do not click on suspicious links in emails, messages, or social media posts. Verify the source before clicking.
- Be careful about what personal information you share publicly — phone numbers, home addresses, and financial details should never be posted online.

These tips were chosen because they are actionable, free, and immediately applicable — a first-year student can implement all four today without any technical expertise.

Section 3: Professional Online Presence

The third section addressed the growing expectation that students and professionals maintain a curated, professional digital identity. The key points included:

- Keep all social media accounts consistent with the image you want to project professionally.
- Create a LinkedIn profile early, even if it only contains your education details.
- Use GitHub to showcase your coding work, projects, and learning progress throughout your degree.
- Always use your college email for academic and professional sign-ups — it signals institutional affiliation and is more trustworthy than a personal email.

This section also included a brief reminder that anything posted online is potentially permanent and publicly accessible, reinforcing the importance of thoughtful digital behaviour.

2.5 Challenges Encountered

While the overall experience of using Canva was positive, I encountered several challenges during the design process:

Challenge 1: Simplifying Complex Concepts

The most significant challenge was distilling complex ideas — particularly the definition of digital literacy and the nuances of safe internet practices — into short, punchy bullet points that could be read in under thirty seconds. Every word in an infographic needs to earn its place. I found myself writing long explanations and then having to cut them down significantly, which is a skill in itself.

This process taught me that effective visual communication is as much about what you omit as what you include. A cluttered infographic with too much text defeats its own purpose — it becomes just a document in disguise.

Challenge 2: Maintaining Visual Hierarchy

A well-designed infographic guides the reader's eye through the content in a logical order. Achieving this visual hierarchy — making the most important information stand out without making the design feel chaotic — required several iterations. I had to experiment with font sizes, colours, and spacing before arriving at a layout that felt balanced.

Challenge 3: Icon Selection

Canva's icon library is vast, which sounds like an advantage but can be overwhelming. Choosing icons that were thematically appropriate, visually consistent in style, and not too literal or too abstract for each section required careful thought. I ultimately chose line-art style icons that matched the overall clean aesthetic of the template.

2.6 Key Learnings

Task 1 yielded several important learnings that extend beyond the immediate deliverable:

- Visual communication is a form of literacy: Just as writing clearly requires practice, designing clear infographics requires thoughtfulness about structure, hierarchy, and audience.
- Simplicity is powerful: The most impactful infographics are the ones that say the most with the fewest words. Economy of language is a skill worth developing.
- Free tools are genuinely capable: Canva demonstrated that professional-quality design is accessible to everyone, regardless of budget or technical background.

This democratisation of design is itself a feature of the digital world worth understanding.

- Content planning precedes design: Starting with a clear content plan before touching the design tool saved significant time and resulted in a more coherent final product.
- Digital tools reduce barriers to creation: A generation ago, creating a professional-looking infographic would have required either specialist software (like Adobe Illustrator) or a graphic designer. Today, a first-year student can produce comparable results in under two hours using a free browser-based tool.

2.7 Official Reflection Write-Up (150–200 Words)

For this task, I used Canva, a free browser-based graphic design platform, to create a one-page Digital Literacy Awareness Infographic. Canva's drag-and-drop interface made it accessible even without any prior design experience, which aligned perfectly with the spirit of this assignment.

The infographic covers three core topics: What Digital Literacy Is, Safe Internet Practices, and Professional Online Presence. The design uses a clean, colour-coded layout with icons and short bullet points to make the information easy to scan at a glance. Each section is visually distinct so that a first-year student can quickly understand the key ideas without feeling overwhelmed by text.

One thing I found particularly interesting was deciding how to simplify complex ideas — such as what it means to maintain a professional online presence — into just two or three bullet points without losing the core message. It taught me that good design is as much about what you leave out as what you include. This exercise made me more conscious of how I present information digitally, which is itself a core digital literacy skill.

2.8 Extended Analysis of Digital Literacy Concepts

Having completed the infographic, this section offers a deeper analysis of the three concepts covered, drawing on academic definitions, real-world examples, and their relevance to the Indian engineering student context.

2.8.1 Defining Digital Literacy: Beyond Basic Computer Skills

Digital literacy is frequently confused with digital fluency or basic computer skills — the ability to type, browse the internet, or use a word processor. But the concept is

substantially richer. Scholars define digital literacy as encompassing several distinct competencies:

- Information literacy: The ability to find, evaluate, and use information from digital sources critically.
- Media literacy: The ability to understand and critically assess digital media content, including social media posts, news articles, and online videos.
- Communication literacy: The ability to communicate clearly, professionally, and ethically across digital channels — email, messaging, social media, and collaborative platforms.
- Technical literacy: A working understanding of how digital tools and systems function, sufficient to use them effectively without needing to be an engineer or programmer.
- Data literacy: The ability to read, interpret, and work with data — increasingly important in a data-driven world.

The infographic focused primarily on the first, third, and fourth of these competencies — information literacy (through safe internet practices), communication literacy (through professional online presence), and an introduction to the concept itself. A complete treatment of digital literacy would require covering all five, but for a one-page infographic aimed at first-year students, the chosen scope was appropriate.

2.8.2 Safe Internet Practices: A Deeper Look

The infographic covered four safe internet practices. Each deserves a more thorough explanation:

Strong Passwords and Password Managers

A strong password is typically at least 12 characters long, contains a mix of uppercase and lowercase letters, numbers, and symbols, and does not contain easily guessable information like names, birthdays, or dictionary words. The challenge with strong passwords is memorability — the stronger the password, the harder it is to remember. Password managers (such as Bitwarden, which is free and open-source) solve this problem by securely storing passwords and auto-filling them on websites. Students should use a different password for every account so that a breach of one account does not compromise all others.

Two-Factor Authentication (2FA)

Two-factor authentication adds a second layer of security beyond the password. Even if an attacker obtains a user's password — through phishing, data breach, or guessing — they cannot access the account without the second factor, which is

typically a one-time code sent via SMS, generated by an authenticator app, or provided by a physical security key. For college students, enabling 2FA on email, UPI apps, and social media accounts is one of the single most impactful security improvements they can make.

Phishing Awareness

Phishing is the practice of deceiving individuals into revealing sensitive information — such as passwords, bank details, or OTPs — by disguising malicious communications as legitimate ones. Phishing attacks arrive via email, SMS (smishing), social media messages, and even phone calls (vishing). Common warning signs include: unexpected urgency ('Your account will be closed in 24 hours'), mismatched sender addresses, suspicious links that don't match the supposed sender's domain, and requests for personal information that a legitimate organisation would never ask for via email.

Personal Information Hygiene

Personal information shared online — phone numbers, addresses, dates of birth, financial details — can be aggregated by malicious actors to commit identity theft, social engineering attacks, or direct financial fraud. Students should audit their social media privacy settings regularly, avoid sharing location information in real time, and be cautious about filling in online forms that request more information than strictly necessary.

2.8.3 Professional Online Presence: Why It Matters for Engineers

The concept of a professional online presence may seem premature for a first-year student who has just begun their degree. In reality, it is never too early. Recruiters, internship coordinators, and academic collaborators routinely search for candidates online before making decisions. A well-maintained GitHub profile demonstrating consistent learning and project work, a complete LinkedIn profile showing academic progress, and a professional email address collectively signal that a student is organised, forward-thinking, and digitally aware.

The infographic summarised this in four bullet points. The extended analysis adds the following context:

- GitHub as a portfolio: Unlike a resume, a GitHub profile is dynamic and verifiable. Potential employers can see not just what a student claims to have built, but the actual code, commit history, and problem-solving approach.
- LinkedIn as a networking tool: LinkedIn is not just a digital resume — it is a professional networking platform. Following industry leaders, joining student groups, and engaging with relevant content builds a student's professional network even before they graduate.
- The permanence of the digital footprint: Content shared online — whether a Tweet, an Instagram post, a comment on a forum, or a LinkedIn post — can remain accessible long after it was originally posted. Students should be aware that future employers, university admissions committees, and professional contacts may encounter this content.
- Separation of personal and professional: Maintaining a distinction between personal and professional online identities is a practical strategy. Using a college email for academic sign-ups, keeping personal social media accounts private, and reserving LinkedIn for professional content are examples of this separation in practice.

3. Task 2 – Build Your Student Digital Portfolio

Module 2 | 20 Marks | Deliverable: Screenshots in task-2-portfolio/ folder + 150–200 word reflection in project report.

3.1 Task Overview and Requirements

Task 2 required setting up a professional digital portfolio across at least three platforms from the following list: GitHub, LinkedIn, Kaggle, Stack Overflow, and ResearchGate. The specific requirements included:

- Creating or updating accounts on at least three of the listed platforms.
- On GitHub: creating a profile README (a special repository named after the username) with at least a name, branch, year, and one sentence about what the student hopes to learn.
- On LinkedIn: filling in the Education section with the current degree, institution, and expected graduation year.
- Taking a screenshot of each profile and adding them to the task-2-portfolio/ folder.
- Writing a 150–200 word reflection on which platforms were chosen, what each is used for, and how they will be used over the next four years.

The task explicitly noted that no photo or contact details were required, that the college email should be used for sign-ups, and that all profiles should be professional and minimal at this stage.

3.2 Platform Selection Rationale

I chose GitHub, LinkedIn, and Kaggle as my three platforms. The reasoning for each selection is detailed below.

Why GitHub?

GitHub is the world's largest platform for hosting, sharing, and collaborating on code. For a Computer Science student, GitHub is not optional — it is the primary platform through which technical work is shared, reviewed, and evaluated. Beyond hosting code, GitHub serves as a version control system (using Git), a collaboration platform for team projects, a portfolio of technical skills, and a community for open-source contribution.

The profile README feature — a special repository named after the GitHub username — appears directly on the user's GitHub profile page and serves as a first

impression for anyone who views the profile. For a first-year student, keeping it simple and honest is the right approach: name, degree, institution, year, and a brief statement of learning intent.

Why LinkedIn?

LinkedIn is the world's largest professional networking platform with over one billion members globally. While it may feel premature to maintain a LinkedIn profile as a first-year student, the platform offers several immediate benefits:

- Connecting with faculty, seniors, and industry professionals in the CSE field.
- Following companies of interest to stay updated on their research, internship postings, and culture.
- Joining student groups and university alumni communities.
- Beginning to build a public record of academic achievements and extracurricular activities that can grow over four years.

Setting up the Education section early ensures that the profile is correctly linked to VIT Bhopal University's institutional page on LinkedIn, which can surface the profile in searches by campus recruiters.

Why Kaggle?

Kaggle is the world's largest data science and machine learning community, owned by Google. While it is most widely known for hosting machine learning competitions, Kaggle also offers free online courses, datasets for practice, notebooks for running code in the cloud, and a public profile that showcases a user's competition rankings, completed courses, and notebooks.

For a CSE student, Kaggle is valuable from the very first year because:

- It provides a safe, structured environment for practising Python and data analysis without needing to set up any local environment.
- Completing Kaggle's free 'Intro to Programming' or 'Python' courses earns certificates that can be added to a LinkedIn profile.
- A Kaggle profile with completed courses and notebooks serves as an additional portfolio component alongside GitHub.
- The platform's community aspect — discussions, notebooks shared by others, and competition write-ups — is an excellent informal learning resource.

Platforms Not Selected and Why

Stack Overflow was not selected primarily because meaningful participation on the platform requires a level of technical expertise that is better developed over the

course of the degree rather than in the first semester. Premature account creation without meaningful contribution does not add value to a portfolio.

ResearchGate was not selected because it is primarily designed for academic researchers — typically postgraduate students, faculty, and professionals who have published papers. A first-year undergraduate student is unlikely to have relevant publications or research output to share on the platform.

3.3 GitHub Profile Setup

3.3.1 Account Creation

I visited github.com and created an account using my college email address (22bce1001@vitbhopal.ac.in). I chose the username rahul-vit-bce to clearly identify myself as a VIT student while keeping the username professional and memorable. After verifying my email, I completed the account setup and began exploring the platform.

3.3.2 Creating the digital-literacy-project Repository

Following the instructions in Appendix A of the project brief, I created a new public repository named digital-literacy-project. I checked the option to 'Add a README file' during repository creation, which automatically created a README.md file that I could edit directly in the browser.

I then edited the README.md to include the project overview, my name and registration number, module summaries, and links to each task folder. The README serves as the front page of the repository — the first thing anyone sees when they visit the repository link.

3.3.3 Creating the Profile README

In addition to the project repository, I created a special repository named rahul-vit-bce (matching my GitHub username exactly). GitHub recognises this as a profile README repository and displays its contents directly on the profile page.

The profile README included the following elements:

- My name: Rahul Kumar Sharma
- Branch and year: B.Tech CSE, First Year, VIT Bhopal University
- A brief learning statement: 'I am a first-year Computer Science student exploring the fundamentals of programming, data structures, and digital tools. I hope to use this space to document my learning journey over the next four years.'

- A note about the Digital Literacy project with a link to the digital-literacy-project repository.
- A simple list of current focus areas: Python basics, Data Structures, Digital Literacy, and Web Fundamentals.

Keeping the profile README minimal and honest at this stage reflects the principle that authenticity is more valuable than an inflated profile. As skills grow, the README will grow with it.

3.3.4 Repository Folder Structure

I created the following folder structure in the digital-literacy-project repository by uploading files to each folder path:

- README.md – Project overview and navigation.
- report/ – Contains this project report.
- task-1-presentation/ – Contains the infographic image and a brief description file.
- task-2-portfolio/ – Contains screenshots of all three platform profiles.
- task-3-platforms/ – Contains coding challenge screenshot and Google Form/Sheet screenshots.
- task-4-email-etiquette/ – Contains email drafts and the social media checklist.
- task-5-cybercrime/ – Contains the case study and prevention checklist.

3.3.5 Understanding Git and Version Control

While this project used GitHub primarily as a file storage and portfolio platform (editing files directly in the browser), it is worth understanding the underlying technology. GitHub is built on Git, a distributed version control system created by Linus Torvalds in 2005.

Version control allows developers to:

- Track every change made to a codebase over time.
- Revert to an earlier version of a file if something goes wrong.
- Work on different features simultaneously using branches.
- Collaborate with others on the same codebase without overwriting each other's work.

Each time a file is saved on GitHub, a 'commit' is created — a snapshot of the file at that point in time, with a message describing what was changed and why. This commit history becomes a transparent record of a developer's working process, which is one of the reasons GitHub profiles are so valued by employers.

3.4 LinkedIn Profile Setup

3.4.1 Account Creation

I visited linkedin.com and created a new account using my college email address. I chose to keep the profile private for now, with only my education details filled in, consistent with the instruction to keep all profiles professional and minimal at this stage.

3.4.2 Education Section

I completed the Education section with the following information:

- School/Institution: VIT Bhopal University
- Degree: Bachelor of Technology (B.Tech)
- Field of Study: Computer Science and Engineering
- Start Year: 2025
- End Year (Expected): 2029
- Grade: N/A (first year, no grades yet to report)
- Activities and Societies: Digital Literacy Club, Student Digital Ambassador Programme

Adding the institution name as 'VIT Bhopal University' automatically linked the profile to the university's official LinkedIn page, making the profile discoverable through the university's alumni and current student network.

3.4.3 About Section

I wrote a brief About section to provide context for anyone who views the profile:

'First-year B.Tech Computer Science student at VIT Bhopal University. Interested in software development, data science, and responsible digital citizenship. Currently building foundational skills in programming and digital literacy. Open to internship opportunities from 2027 onwards.'

This About section is honest, forward-looking, and includes a subtle indication of availability for internships — without making promises about qualifications that do not yet exist.

3.4.4 LinkedIn as a Long-Term Tool

LinkedIn's value compounds over time. A profile created in the first year, regularly updated with academic achievements, projects, certifications, and extracurricular activities, becomes a rich record of professional development by the time a student is applying for full-time roles. The platform's algorithm also rewards consistent engagement — posting thoughtful comments, sharing relevant articles, and connecting with peers and faculty increases profile visibility.

Over the next four years, I plan to add the following to my LinkedIn profile:

- Internship experiences (from second year onwards).
- Technical certifications from platforms like Coursera, edX, and NPTEL.
- Projects completed during the degree, with links to GitHub repositories.
- Research papers or academic publications, if applicable.
- Recommendations from faculty and internship supervisors.

3.5 Kaggle Profile Setup

3.5.1 Account Creation

I visited kaggle.com and created a free account using my college email address. Kaggle allows sign-up with Google, which simplified the process. After completing the account setup, I was taken to the Kaggle home feed — a personalised view of competitions, notebooks, and discussions relevant to my stated interests (programming and data science).

3.5.2 Profile Setup

I completed the basic profile information, including:

- Display name: Rahul Kumar Sharma
- Occupation: Student
- Organisation: VIT Bhopal University
- Location: Bhopal, Madhya Pradesh, India
- Bio: 'First-year B.Tech CSE student exploring data science and programming fundamentals.'

I kept the profile minimal and professional, without adding any social media links or personal contact information.

3.5.3 Exploring Kaggle's Learning Resources

After setting up the profile, I explored Kaggle's free course offerings. The platform offers courses in:

- Intro to Programming (Python-based, no prior experience required)
- Python
- Intro to Machine Learning
- Data Visualisation
- SQL
- Natural Language Processing

These courses are self-paced, free, and result in certificates that can be displayed on the Kaggle profile and shared on LinkedIn. For a first-year CSE student, the Intro

to Programming and Python courses are the natural starting point and directly complement the programming coursework in the B.Tech curriculum.

3.6 Official Reflection Write-Up (150–200 Words)

For this task, I set up professional profiles on three platforms: GitHub, LinkedIn, and Kaggle. Each serves a distinct but complementary purpose in building my digital portfolio.

GitHub is the world's most widely used platform for hosting and sharing code. I created both a project repository (digital-literacy-project) and a profile README that introduces me, my branch, my year, and my learning goals. Over the next four years, GitHub will serve as a living record of every project I build, every algorithm I implement, and every collaborative exercise I participate in.

LinkedIn is the primary professional networking platform used by employers and recruiters. I completed my Education section with my current degree, institution, and expected graduation year. As I progress through my degree, I plan to add internships, certifications, and projects.

Kaggle is Google's data science and machine learning community. I created a profile and explored its free course offerings in Python and data science. I plan to use Kaggle to practise programming, complete structured courses, and eventually participate in competitions.

Together, these three platforms form the foundation of a professional digital identity that I will build upon throughout my four-year degree.

3.7 Extended Analysis: Building a Digital Identity

3.7.1 The Digital Identity Paradox

There is a paradox at the heart of building a digital identity as a student: the best time to start is before you have anything impressive to show, because the practice of maintaining a digital portfolio is itself a skill that takes time to develop. Many students make the mistake of waiting until the final year — or until placement season — to create their LinkedIn profile or GitHub account. By that point, they are scrambling to document four years of work from memory, rather than maintaining a running record.

Starting early, even with minimal content, has several advantages:

- Profile age matters: Older, established profiles signal consistency and commitment.
- Incremental documentation: Recording each project, certification, and achievement as it happens is easier and more accurate than reconstructing it later.
- Network effects: Professional connections made early — with faculty, seniors, and industry professionals encountered at events — accumulate over time.
- Habit formation: The habit of maintaining a professional digital presence is easier to form early than to retrofit later.

3.7.2 Privacy Considerations in Digital Portfolio Building

Building a public digital portfolio involves trade-offs between visibility and privacy.

The following guidelines help navigate these trade-offs:

- Use a college email for sign-ups: This is already mandated by the project brief and is good practice for academic and professional accounts.
- Avoid sharing personal contact details publicly: Phone numbers, home addresses, and personal email addresses should not appear on public profiles.
- Review privacy settings regularly: LinkedIn, GitHub, and Kaggle all offer granular privacy controls. Understanding and using these settings gives control over who can see what.
- Be consistent across platforms: Using the same professional name across all platforms makes it easier for the right people to find you and harder for bad actors to impersonate you.

3.7.3 The Four-Year Digital Portfolio Plan

The following is a rough roadmap for how the three chosen platforms will be used over the course of the B.Tech degree:

Year 1 (Current):

- GitHub: Profile README, digital-literacy-project repository, first programming exercises.
- LinkedIn: Education section, About section, connection with classmates and faculty.
- Kaggle: Account setup, Intro to Programming course, exploration of the platform.

Year 2:

- GitHub: Data structures and algorithms practice repositories, first collaborative projects.
- LinkedIn: First internship (if secured), technical certifications (NPTEL, Coursera).
- Kaggle: Completion of Python and Intro to Machine Learning courses, first notebooks.

Year 3:

- GitHub: Major project repositories, open-source contributions, internship projects.
- LinkedIn: Research projects, conference attendance, peer recommendations.
- Kaggle: Competition participation, public notebooks, data science portfolio.

Year 4:

- GitHub: Capstone project, comprehensive portfolio of four years of work.
- LinkedIn: Full professional profile ready for job applications and campus placements.
- Kaggle: Competition rankings, certificates in multiple data science domains.

4. Task 3 – Explore Coding and Collaboration Platforms

Module 3 | 20 Marks | Deliverable: Screenshots in task-3-platforms/ + 150–200 word reflection in project report.

4.1 Task Overview and Requirements

Task 3 required building familiarity with two categories of digital platforms: coding practice platforms and cloud collaboration tools. The task was divided into two parts:

Part A – Coding Practice (choose one platform):

- Create an account on HackerRank, CodeChef, or HackerEarth.
- Complete at least one beginner-level challenge or skill test.
- Take a screenshot of the completed challenge and badge/score, and add to task-3-platforms/.

Part B – Google Workspace Collaboration (mandatory):

- Create a Google Form with exactly 5 questions for a Digital Literacy Awareness Quiz.
- Include at least one multiple-choice question and one short-answer question.
- Share the form link in the README and add screenshots of the form and linked Google Sheet to task-3-platforms/.

4.2 Part A: Coding Practice Platform

4.2.1 Platform Selection: HackerRank

I selected HackerRank as my coding practice platform. HackerRank is one of the most widely used technical hiring platforms in the world, used by companies including Google, Amazon, Microsoft, and thousands of others to assess candidates. For students, HackerRank offers structured learning paths, skill certifications, and a competitive community.

The reasons for choosing HackerRank over CodeChef and HackerEarth were:

- Interface clarity: HackerRank's problem interface is clean and well-documented, with clear problem statements and examples.
- Beginner friendliness: The platform offers a dedicated 'Prepare' section with problems organised by domain (Python, Java, SQL, etc.) and difficulty level.
- Certification value: HackerRank's skill certifications (e.g., Python Basic, SQL Basic) are recognised by employers and can be displayed on LinkedIn.

- Global recognition: HackerRank's brand is more widely recognised by Indian employers than HackerEarth, and it is used heavily in campus recruitment.

4.2.2 Account Creation on HackerRank

I visited hackerrank.com and created a free account using my college email address. The registration process required me to:

13. Enter my name, email, and a password.
14. Verify my email address by clicking a confirmation link.
15. Complete a brief onboarding survey about my skill level and areas of interest (I selected 'Beginner' and 'Python' as primary interests).
16. Optionally set up a public profile, which I configured with my name and a brief bio.

After completing registration, I was directed to the HackerRank dashboard, which showed recommended challenges and learning paths based on my survey responses.

4.2.3 The 'Solve Me First' Challenge

I chose to complete the 'Solve Me First' challenge from the HackerRank Algorithms section. This challenge is specifically designed as an introduction to the HackerRank platform — it helps new users understand the problem format, input/output structure, and submission process.

Problem Statement:

Complete the function solveMeFirst to compute the sum of two integers. The function receives two integers as parameters, a and b, and must return their sum.

My Approach:

The problem was intentionally trivial — it was not designed to test algorithmic thinking but to familiarise users with the platform. Using HackerRank's built-in code editor, I selected Python 3 as my language and wrote the following solution:

```
def solveMeFirst(a, b):
    return a + b

a = int(input())
b = int(input())
print(solveMeFirst(a, b))
```

I clicked 'Run Code' to test the solution against the sample test cases, which passed successfully. I then clicked 'Submit' to submit the final solution. The submission was accepted with a score of 10/10.

4.2.4 Python Introduction Challenge

In addition to the Solve Me First challenge, I also completed the 'Python: Division' challenge from the Python domain. This challenge required printing both the integer division and float division of two numbers.

My solution:

```
a = int(input())
b = int(input())
print(a // b)
print(a / b)
```

This challenge was also accepted successfully, and completing it contributed to my progress in the Python domain on HackerRank.

4.2.5 HackerRank as a Long-Term Learning Tool

Beyond the beginner challenges completed for this task, HackerRank offers a structured learning path that I intend to continue using throughout my degree. The platform's problem library is organised into the following domains, each of which maps to core CSE curriculum areas:

- Algorithms: Sorting, searching, dynamic programming, graph theory, and more.
- Data Structures: Arrays, stacks, queues, linked lists, trees, and graphs.
- Mathematics: Number theory, combinatorics, probability, and linear algebra.
- Artificial Intelligence: Bot programming, machine learning basics.
- SQL: Database querying and manipulation.
- Python, Java, C++: Language-specific challenges.

HackerRank's certification programme is particularly valuable. Certifications in Python Basic, SQL Basic, and Problem Solving (Basic) are free, proctored online assessments that result in verifiable certificates displayed on the HackerRank profile and shareable on LinkedIn. These certificates carry weight with Indian tech employers because HackerRank is the platform many of them use for technical screening.

4.2.6 Comparison: HackerRank vs CodeChef vs HackerEarth

For context, here is a brief comparison of the three suggested platforms:

HackerRank:

- Best for: Interview preparation, skill certifications, domain-specific practice.
- Strengths: Clean interface, well-structured learning paths, industry recognition.
- Weaknesses: Fewer competitive programming contests than CodeChef.

CodeChef:

- Best for: Competitive programming, monthly contests, college teams.
- Strengths: Strong Indian community, regular rated contests, ICPC preparation.
- Weaknesses: Interface can feel less intuitive for beginners; problem quality can vary.

HackerEarth:

- Best for: Company-specific hiring challenges, sprint competitions.
- Strengths: Used by many Indian startups for hiring.
- Weaknesses: Smaller global community than HackerRank; fewer structured learning paths.

4.3 Part B: Google Workspace Collaboration

4.3.1 Why Google Forms?

Google Forms is a free, browser-based form builder included in the Google Workspace suite. It allows users to create surveys, quizzes, and questionnaires with a wide variety of question types — multiple choice, short answer, paragraph, checkboxes, linear scale, and more. Responses are automatically collected in a linked Google Sheet, making analysis straightforward.

For the Student Digital Ambassador role, a Google Form quiz on Digital Literacy is a genuinely useful deliverable: it can be shared with batchmates via a link, completed on any device (phone, tablet, or laptop), and the results can be reviewed in real time through the linked Google Sheet.

4.3.2 Creating the Digital Literacy Awareness Quiz

I accessed Google Forms through my college Google account (which provides access to the full Google Workspace suite via my VIT Bhopal institutional account). I created a new blank form and titled it: 'Digital Literacy Awareness Quiz – Student Digital Ambassador Programme'.

I added a brief description: 'This short quiz tests your awareness of digital literacy concepts, safe internet practices, and professional online tools. It should take no more than 5 minutes to complete. All responses are anonymous.'

The five questions I created are as follows:

Question 1 (Multiple Choice):

Title: 'Which of the following best describes Digital Literacy?'

- A. The ability to type quickly on a computer
- B. The ability to access, evaluate, use, and create information using digital tools
- C. The ability to code in Python or Java
- D. The ability to use social media platforms

Correct Answer: B

Rationale: This question tests the core definition of digital literacy at a conceptual level.

Question 2 (Multiple Choice):

Title: 'Which of the following is a safe internet practice?'

- A. Using the same password for all your accounts
- B. Clicking on links in emails from unknown senders
- C. Enabling two-factor authentication on your accounts
- D. Sharing your phone number on public social media profiles

Correct Answer: C

Rationale: This question tests awareness of specific, actionable security practices.

Question 3 (Multiple Choice):

Title: 'What is the primary purpose of GitHub for a CSE student?'

- A. To watch coding tutorial videos
- B. To store and share code, manage version control, and build a technical portfolio
- C. To find internship opportunities
- D. To participate in online coding contests

Correct Answer: B

Rationale: This question assesses familiarity with GitHub's core function.

Question 4 (Short Answer):

Title: 'In your own words, describe one thing you can do right now to improve your online safety.'

Type: Short Answer (free text)

Rationale: This open-ended question encourages reflective thinking and cannot be answered by guessing — it requires actual understanding.

Question 5 (Short Answer):

Title: 'Name one platform (other than WhatsApp or Instagram) that you plan to use for your professional or academic growth, and explain why.'

Type: Short Answer (free text)

Rationale: This question encourages students to think about platforms they are not yet using and articulate a reason for choosing one, promoting reflective digital behaviour.

4.3.3 Linking to Google Sheets

After setting up all five questions, I accessed the Responses tab in Google Forms and clicked the green Sheets icon to create a linked Google Sheet. Google automatically created a new spreadsheet titled 'Digital Literacy Awareness Quiz (Responses)' and linked it to the form. Every new form submission automatically adds a new row to this spreadsheet, with columns for the timestamp and each question's response.

I took a screenshot of the Google Form (in preview mode, showing all five questions) and a screenshot of the linked Google Sheet (response view, showing the column headers and one test response I submitted myself). Both screenshots were uploaded to the task-3-platforms/ folder in the GitHub repository.

4.3.4 The Form Link

The shareable link to the Digital Literacy Awareness Quiz was added to the README.md of the GitHub repository, under the Task 3 section, as required by the project brief. The link format is: https://forms.gle/[unique-form-id]

As the Student Digital Ambassador, sharing this form link with batchmates via WhatsApp, email, or the class group would allow real data collection and create a genuine awareness initiative — which was exactly the spirit of the task.

4.3.5 Google Workspace Beyond Forms

While Google Forms was the specific tool used in this task, it is worth contextualising it within the broader Google Workspace ecosystem, which is widely used in academic and professional settings:

- Google Docs: Real-time collaborative word processing, equivalent to Microsoft Word.
- Google Sheets: Cloud-based spreadsheet application, equivalent to Microsoft Excel.
- Google Slides: Cloud-based presentation tool, equivalent to Microsoft PowerPoint.

- Google Drive: Cloud storage for all Google Workspace files, accessible from any device.
- Google Meet: Video conferencing integrated with Google Calendar and Gmail.
- Google Calendar: Scheduling and calendar management, shareable with others.

For a CSE student, proficiency in the Google Workspace suite is a practical skill that will be expected in virtually every academic collaboration, internship, and workplace setting. The ability to set up a shared Google Drive folder, co-author a Google Doc, and analyse data in Google Sheets are not advanced technical skills — they are baseline expectations.

4.4 Official Reflection Write-Up (150–200 Words)

For Part A of this task, I chose HackerRank as my coding practice platform. After creating an account with my college email, I completed two beginner-level challenges: 'Solve Me First' (which introduced the platform's problem format and submission process) and 'Python: Division' (which tested basic Python syntax). Both were submitted successfully and earned a score of 10/10.

For Part B, I created a five-question Digital Literacy Awareness Quiz using Google Forms. The quiz includes three multiple-choice questions and two short-answer questions, covering the definition of digital literacy, safe internet practices, and platform awareness. The form is linked to a Google Sheet that automatically collects and organises responses.

Both platforms will support my academic journey in different ways. HackerRank provides a structured environment for practising programming skills and preparing for technical interviews, while Google Workspace offers essential collaboration tools that are used in virtually every academic and professional context. Learning to use these platforms competently from the first year of my degree gives me a meaningful head start.

4.5 Extended Analysis: The Role of Collaborative Tools in Engineering Education

4.5.1 The Shift to Cloud-Based Collaboration

The COVID-19 pandemic of 2020–2022 accelerated a shift in how teams — academic and professional alike — collaborate on shared work. The forced transition to remote work and online education normalised cloud-based collaboration tools:

Google Workspace, Microsoft 365, Zoom, Slack, and GitHub. By 2025, these tools are not emergency measures but standard infrastructure.

For engineering students, this shift has significant implications. Collaborative coding, shared design documents, real-time peer review, and remote project management are now expected skills. Understanding how to use these tools — not just superficially but confidently and efficiently — is part of what it means to be a digitally literate engineer in the modern era.

4.5.2 Coding Platforms and Skill Development

Coding practice platforms like HackerRank, CodeChef, LeetCode, and Codeforces serve multiple functions in a CSE student's learning journey:

Structured practice:

The curriculum of a B.Tech programme covers theory extensively but may not provide sufficient practice in applying that theory through coding. Dedicated coding platforms fill this gap by offering thousands of curated problems organised by topic and difficulty.

Assessment and benchmarking:

A student who has solved 100 HackerRank problems and earned a Python Basic certification has objective evidence of their coding competence. This is more credible to an employer than a self-reported claim of 'proficiency in Python' on a resume.

Interview preparation:

Technical interviews at most product-based companies (Google, Amazon, Microsoft, Flipkart, Swiggy, etc.) involve live coding sessions where candidates are asked to solve algorithmic problems. Practising on platforms like HackerRank or LeetCode from the first year of the degree, even at a beginner level, establishes the habit of regular practice early.

Community and exposure:

The competitive programming community on platforms like CodeChef and Codeforces exposes students to a global community of programmers, contest problems from companies, and editorial write-ups that explain elegant solutions in detail.

4.5.3 Google Forms as a Data Collection Tool

The Google Form created for this task is not merely an academic exercise — it is a functional data collection instrument. When shared with batchmates, it can:

- Measure baseline digital literacy awareness among first-year students.
- Identify specific areas where awareness is low (e.g., if most respondents cannot correctly identify safe internet practices).
- Provide data for a follow-up presentation or workshop by the Student Digital Ambassador.
- Demonstrate to faculty and peers that the ambassador role is producing real, actionable outputs.

This connection between the tool and its real-world application — using a Google Form not just to fulfil an assignment requirement but to actually gather useful data — is an example of what digital literacy looks like in practice.

4.5.4 Digital Collaboration Best Practices

Working with cloud-based collaboration tools requires awareness of certain best practices:

- Naming conventions: Files, forms, and documents should be named clearly and consistently (e.g., 'Task3_DigitalLiteracyQuiz_22BCE1001' rather than 'Untitled form').
- Access control: When sharing Google Forms, Sheets, or Docs, consider who should have view-only access versus edit access. For a survey, respondents should have view access to the form only, not the response sheet.
- Data privacy: When collecting responses through a Google Form, inform respondents whether the data is anonymous, how it will be used, and whether it will be shared with others.
- Version history: Google Docs and Sheets maintain a version history of all edits. This is a valuable safety net — if content is accidentally deleted, it can be recovered. Understanding this feature prevents panic in collaborative situations.
- Offline access: Google Workspace documents can be configured for offline access, ensuring work can continue without an internet connection and syncing when connectivity is restored.

5. Task 4 – Professional Email Etiquette and Communication

Module 4 | 20 Marks | Deliverable: Email drafts + social-media-checklist in task-4-email-etiquette/ + 100–150 word reflection in project report.

5.1 Task Overview and Requirements

Task 4 addressed one of the most practically important dimensions of digital literacy: professional communication. The task was divided into two parts:

Part A – Write two professional emails:

- Email 1: A student to a professor, requesting an extension on an assignment submission deadline.
- Email 2: A student to an internship coordinator, expressing interest in a summer internship.
- Both emails must include a clear subject line, professional greeting, well-structured body, and proper sign-off.

Part B – Create a Social Media Do's and Don'ts checklist:

- At least 5 Do's and 5 Don'ts for responsible social media use as a college student.
- Save the checklist in task-4-email-etiquette/ as social-media-checklist.md or .txt.
- Write a 100–150 word reflection on a situation where poor digital communication caused a real problem.

5.2 Part A: Professional Email Drafts

5.2.1 Principles of Professional Email Writing

Before presenting the email drafts, it is useful to articulate the principles that guided their construction. Professional email writing — whether to a faculty member, an employer, or a colleague — is governed by a set of conventions that signal respect, clarity, and competence.

Subject Line:

The subject line is the most important element of a professional email. It determines whether the email is opened promptly or deprioritised. A good subject line is:

- Specific: It identifies the purpose of the email immediately.
- Concise: It is typically no longer than 6–8 words.
- Professional: It avoids casual language, abbreviations, or excessive punctuation.

- Informative: It gives the recipient enough context to understand the email's urgency and relevance before opening it.

Salutation/Greeting:

The greeting sets the tone for the entire email. For formal communications:

- 'Dear Professor [Last Name],' is appropriate for faculty.
- 'Dear Mr./Ms. [Last Name],' or 'Dear [Full Name],' is appropriate for professional contacts whose gender or title is uncertain.
- Avoid 'Hi', 'Hey', or 'Hello' in formal emails — these are appropriate for familiar colleagues but not for faculty or employers.
- Never start an email with 'Sir/Madam' without a name — it sounds impersonal and generic.

Body:

The body of a professional email should be organised into:

17. Opening sentence: State the purpose of the email immediately.
18. Context/Background: Provide any necessary context briefly.
19. Request or Main Point: State what you are asking for or communicating.
20. Supporting Rationale: Briefly explain why, if relevant.
21. Closing sentence: Thank the recipient for their time and indicate any next steps.

Sign-off:

The sign-off closes the email professionally:

- 'Yours sincerely,' is appropriate when the recipient is named.
- 'Kind regards,' is appropriate for moderately formal emails.
- 'Best regards,' is a widely acceptable, slightly less formal option.
- Avoid 'Thanks and regards', 'Warm regards', or 'Cheers' in formal emails to faculty or employers.

The signature below the sign-off should include: full name, registration number (for academic emails), programme, institution, and contact email.

5.2.2 Email 1: Assignment Extension Request

SUBJECT: Request for Assignment Submission Deadline Extension – CSE2001 Data Structures (Assignment 2)

Dear Professor Sharma,

I hope this email finds you well. I am writing to respectfully request an extension of three days for the submission of Assignment 2 for CSE2001 – Data Structures, which is currently due on 2 April 2026.

I have been making steady progress on the assignment; however, I experienced an unexpected medical issue over the past week that significantly disrupted my study schedule. I was required to visit the university health centre on 25 March and was advised to rest for several days, during which my ability to complete coursework was severely limited. I have attached a copy of the medical certificate from the health centre for your reference.

I understand and respect the importance of submission deadlines, and I assure you that I am not seeking an extension lightly. If granted an extension until 5 April 2026, I am confident that I will be able to complete and submit work of the quality expected in this course.

I would be grateful for the opportunity to discuss this further at your convenience, should you require additional information. Thank you very much for your time and consideration.

Yours sincerely,

Rahul Kumar Sharma

Reg. No.: 22BCE1001

B.Tech – Computer Science and Engineering, Year 1

VIT Bhopal University

Email: 22bce1001@vitbhopal.ac.in

Analysis of Email 1

Subject Line Analysis:

The subject line 'Request for Assignment Submission Deadline Extension – CSE2001 Data Structures (Assignment 2)' is specific, contains the course code and assignment number, and immediately communicates both the nature of the request ('extension') and the relevant context ('CSE2001', 'Assignment 2'). A professor receiving dozens of emails daily can immediately understand the purpose without opening the email.

Structure Analysis:

The email follows a four-part structure: (1) purpose statement, (2) context and evidence, (3) the request with a specific proposed deadline, and (4) an offer to

discuss further. Mentioning the medical certificate preemptively demonstrates accountability and prepares the reader for the attached evidence.

Tone Analysis:

The tone is appropriately formal and respectful without being obsequious. Phrases like 'respectfully request', 'I understand and respect the importance of submission deadlines', and 'I am not seeking an extension lightly' demonstrate that the student acknowledges the professor's authority and the inconvenience of the request.

5.2.3 Email 2: Internship Expression of Interest

SUBJECT: Expression of Interest – Summer Internship 2026 (Computer Science / Software Development)

Dear Ms. Desai,

I am writing to express my keen interest in a summer internship opportunity at TechVenture Solutions Pvt. Ltd. I am currently a first-year B.Tech student in Computer Science and Engineering at VIT Bhopal University (Reg. No. 22BCE1001, expected graduation: 2029) and am eager to supplement my academic learning with practical industry experience during the summer break (May – July 2026).

I came across TechVenture Solutions through a LinkedIn post by one of your team members and was impressed by the company's focus on scalable software solutions for the education sector — an area I find both technically challenging and socially meaningful. I am particularly interested in contributing to your backend development or data analytics teams, where I believe my growing skills in Python and SQL could add value.

While I am at an early stage of my degree, I am a proactive and disciplined learner. I have already completed beginner-level programming challenges on HackerRank, am working through Kaggle's Python course, and have set up a GitHub portfolio to document my learning progress. I am committed to learning quickly and contributing meaningfully, even at an introductory level.

I have attached a brief one-page resume for your consideration. I would be grateful for the opportunity to speak with you or a member of your team about any available internship roles. I am available for a call or video meeting at your convenience.

Thank you for taking the time to consider my application.

Kind regards,

Rahul Kumar Sharma

B.Tech CSE Year 1, VIT Bhopal University

Email: 22bce1001@vitbhopal.ac.in

GitHub: github.com/rahul-vit-bce

LinkedIn: linkedin.com/in/rahul-kumar-sharma-vitbhopal

Analysis of Email 2

Subject Line Analysis:

The subject line specifies both the type of application ('Expression of Interest') and the role category ('Summer Internship 2026, Computer Science / Software Development'). Including the year prevents the email from being misidentified as a previous cycle's application. The specificity helps the internship coordinator file the email appropriately.

Differentiation Strategy:

The email mentions specifically how the applicant discovered the company (LinkedIn post by a team member) and why they are interested in this particular company (education sector focus). This personalisation distinguishes the email from generic mass applications and demonstrates that genuine research was done before writing.

Handling the 'First-Year Problem':

A first-year student applying for an internship faces the challenge of limited experience. The email addresses this directly by acknowledging the early stage of the degree, listing specific evidence of proactive learning (HackerRank, Kaggle, GitHub), and framing the application as introductory-level — setting realistic expectations while demonstrating initiative.

5.3 Part B: Social Media Do's and Don'ts Checklist

The following checklist was saved as social-media-checklist.md in the task-4-email-etiquette/ folder of the GitHub repository.

Social Media Do's for College Students

22. Do keep your public profiles professional: Assume that any post, comment, or photo you make publicly visible can be seen by a future employer, faculty member, or university admissions committee.
23. Do use privacy settings thoughtfully: Review and update the privacy settings on all social media platforms at least once a semester. Make personal content visible only to trusted connections.
24. Do think before you post: Ask yourself — would you be comfortable if your parents, professor, or future employer saw this post? If the answer is no, reconsider posting it.
25. Do engage constructively in online discussions: If you participate in online debates or comment on news articles, maintain a respectful, evidence-based tone. Avoid personal attacks, inflammatory language, and spreading unverified claims.
26. Do credit others' work: If you share, repost, or build upon someone else's content, give them appropriate credit. This applies to academic content, creative work, and ideas.
27. Do use social media for professional networking: Follow industry leaders, researchers, and companies in your field of interest. Engage thoughtfully with their content — this builds your professional network over time.
28. Do maintain a consistent professional identity across platforms: Use the same professional name, a consistent and appropriate profile photo, and a coherent personal brand across LinkedIn, GitHub, and other professional platforms.

Social Media Don'ts for College Students

29. Don't share personal contact information publicly: Phone numbers, home addresses, and personal email addresses should never appear on public social media profiles.
30. Don't post or share content you haven't verified: Sharing misinformation — even unintentionally — can damage your credibility and contribute to real-world harm. Before sharing news, verify it with a reliable source.
31. Don't engage in cyberbullying or online harassment: Bullying, trolling, or targeting individuals with malicious content online is both ethically wrong and potentially illegal under the Information Technology Act, 2000.
32. Don't post during academic or professional events: Tweeting from a lecture, posting Instagram Stories during an internship meeting, or texting during a group project session are all signals of poor professional judgment.
33. Don't share confidential academic or organisational information: If you are part of a project team, a student committee, or an internship, do not post confidential discussions, internal documents, or non-public decisions on social media.
34. Don't use social media as a substitute for direct communication: Important conversations — particularly sensitive, emotional, or conflict-related ones — should happen in person or via a private, direct channel, not on a public post or comment thread.

35. Don't leave old or embarrassing content unaddressed: Periodically audit your social media history. Posts, comments, and photos from earlier years that no longer reflect your values or professional identity should be removed or archived.

5.4 Communication Scenario Analysis

The following scenario illustrates how poor digital communication can have serious real-world consequences:

Scenario: The Premature LinkedIn Post

A final-year B.Tech student at a well-known engineering college had just completed a successful round of campus interviews with a major IT company. Excited about the positive response from the interviewers, she posted on LinkedIn: 'Thrilled to share that I have received a job offer from [Company Name]! So grateful to join the team in July 2024! #placement #excited #blessed'

The post received hundreds of likes and congratulatory comments. However, she had not yet received the official offer letter — only a verbal indication from the interviewer that things looked positive. Two days later, the company's HR department reached out to inform her that, following a final background check, the offer would not proceed.

The student was now in the deeply uncomfortable position of having publicly announced an offer that no longer existed. She had to edit the post, face public embarrassment, and manage the perceptions of her peers, family, and professional network. The company's HR team, meanwhile, saw the premature post as a breach of confidentiality and professional discretion.

What Could Have Been Done Differently:

- She should have waited for the official written offer letter before making any public announcement.
- Even upon receiving the offer, she should have considered messaging close contacts privately before making a public LinkedIn post.
- The post should have been phrased more cautiously even if the offer were confirmed — for example, 'Excited to be starting a new chapter soon!' rather than naming the specific company before the onboarding process is complete.
- She could have asked the company's HR team whether it was appropriate to announce the offer on social media before doing so.

5.5 Official Reflection Write-Up (100–150 Words)

One powerful example of poor digital communication causing a real problem is the case of students prematurely announcing job offers on LinkedIn before receiving official offer letters. This happens frequently during campus placement season, when excitement clouds professional judgment. In one such scenario, a student publicly announced an offer that was subsequently withdrawn following a background check. The result was public embarrassment, reputational damage, and a strained relationship with the company.

The key lesson is that digital communication, especially on public platforms, requires the same level of deliberateness and verification that any formal professional communication would demand. What could have been done differently: wait for the written offer letter, seek permission from the employer before announcing, and phrase any announcement cautiously. In digital communication, patience is a professional virtue.

5.6 Extended Analysis: Digital Communication in the 21st Century

5.6.1 The Evolution of Professional Communication

Professional communication has evolved dramatically over the past three decades. In the 1990s, professional correspondence was primarily conducted via physical letters and fax. Email emerged in the late 1990s and rapidly became the dominant channel for professional communication. By the 2010s, instant messaging platforms (Slack, Microsoft Teams, WhatsApp Business) began supplementing email for internal communication. Today, in the 2020s, professional communication happens across email, messaging platforms, video calls, collaborative documents, and social media — simultaneously.

For a first-year engineering student, understanding this landscape is critical. Each channel has its own norms, expectations, and appropriate use cases:

- Email: Formal, documented, appropriate for external communication, requests, and situations requiring a written record.
- Messaging (Slack/Teams): Informal-to-semi-formal, real-time, appropriate for internal team communication, quick questions, and status updates.
- Video calls (Zoom/Meet): Appropriate for discussions that benefit from non-verbal cues, collaborative problem-solving, and team check-ins.
- LinkedIn: Professional networking, job applications, sharing professional achievements, and engaging with industry content.

- GitHub: Technical collaboration, code review, project management, and open-source contribution.

5.6.2 Email Etiquette: Common Mistakes

The following are the most common email etiquette mistakes made by students:

- Vague subject lines: 'Hi', 'Query', or 'Doubt' are unacceptable subject lines for emails to faculty or employers.
- Omitting a greeting: Beginning an email with the body text, without a salutation, is disrespectful and abrupt.
- Replying to the wrong email: Replying to a group email when only one person needs to receive the response, or hitting 'Reply All' unnecessarily, clutters inboxes and can embarrass the sender.
- Using casual language: 'pls', 'asap', 'lol', or 'btw' have no place in a professional email.
- Forgetting attachments: Mentioning an attached document in the body of the email but forgetting to actually attach it is one of the most common email mistakes.
- Using an inappropriate email address: Sending a professional email from an address like 'cool_rahul_gaming@gmail.com' signals poor professional judgment. Always use a college or professional email address.
- Not proofreading: Spelling errors, grammatical mistakes, and formatting issues in a professional email undermine credibility.

5.6.3 Cross-Cultural Communication Awareness

As CSE graduates increasingly work in global or multinational organisations, awareness of cross-cultural communication differences becomes important. For example:

- In Japanese business culture, email communication tends to be extremely formal, with elaborate greetings and extensive use of honorifics.
- In North American and British professional cultures, email communication is generally more concise and direct, though still professional in tone.
- In German business culture, formal titles are used consistently, and emails are expected to be highly structured and factual.
- In Indian professional culture, email communication often blends formal language with a slightly warmer, more relationship-oriented tone.

Understanding these differences helps avoid miscommunication and builds stronger professional relationships with international colleagues, supervisors, and clients.

5.6.4 Responsible Social Media Use: The Legal Dimension

The Social Media Do's and Don'ts checklist addressed ethical norms. It is also worth understanding the legal framework governing digital communication in India:

- The Information Technology Act, 2000 (and its 2008 amendments) governs online communication, cybercrime, and digital commerce in India. Section 66A, which criminalised offensive online speech, was struck down by the Supreme Court in 2015 (Shreya Singhal v. Union of India). However, other provisions — particularly those related to cyberbullying, online harassment, and impersonation — remain in force.
- Section 67 of the IT Act criminalises the publication of obscene material in electronic form.
- Section 499/500 of the Indian Penal Code (IPC) covers defamation, which can apply to online posts and comments that damage a person's reputation.
- Section 354D IPC covers stalking, including online stalking (cyberstalking).

Students who engage in cyberbullying, online harassment, or defamation through social media can face both institutional consequences (disciplinary action from the university) and legal consequences (police complaints, FIRs, and prosecution under the IT Act and IPC).

6. Task 5 – Cybercrime Awareness Case Study and Prevention Guide

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Module 5 | 20 Marks | Deliverable: casestudy.md + prevention-checklist.md in task-5-cybercrime/ + 100–150 word reflection in project report.

6.1 Task Overview and Requirements

Task 5 required researching a cybercrime scenario and producing an awareness resource. The task was divided into two parts:

Part A – Case Study:

- Choose one cybercrime type: Phishing, UPI/Online Payment Fraud, Identity Theft, Cyberbullying, or Ransomware.
- Write a 200–250 word case study covering: what the crime is, how it typically happens, who is targeted, and the consequences.
- May use a real published incident or a realistic fictional scenario.

Part B – Prevention Checklist:

- Create a 'Stay Safe Online' checklist with at least 8 specific, actionable tips for college students in India.
- At least two tips must be specific to UPI/financial safety.
- Include reporting channel: cybercrime.gov.in and Helpline 1930.
- Write a 100–150 word reflection on what surprised you most and one habit you will personally change.

6.2 Part A: Case Study – UPI/Online Payment Fraud

6.2.1 Why UPI Fraud?

I selected UPI/Online Payment Fraud as my cybercrime topic because it is the most directly relevant form of cybercrime for college students in India in 2025. India's UPI ecosystem processed over 17 billion transactions worth more than ₹24 trillion in a single month in late 2024 — the largest real-time payment volume of any country in the world. With this scale comes a corresponding increase in fraud attempts targeting UPI users.

First-year college students are particularly vulnerable because:

- They are new to financial independence, often managing their own bank accounts and UPI apps for the first time.

- They may be unfamiliar with the technical mechanisms of UPI, making them susceptible to social engineering.
- They are frequently online and active on platforms (WhatsApp, Telegram, Instagram) where UPI fraud links and scam messages circulate.
- Financial pressure — fees, hostel expenses, textbook costs — makes offers of 'easy money' or 'discounts' psychologically appealing.

6.2.2 The Case Study (Official 200–250 Words)

Case Study: The 'Payment Received' Fraud – A UPI Scam Targeting College Students

Vikram is a first-year B.Tech student living in a college hostel in Bhopal. To supplement his monthly allowance, he listed his old laptop for sale on OLX, a popular online classifieds platform, for ₹18,000.

A buyer contacted him via the platform, expressing immediate interest. The buyer said he was in the Indian Army and stationed in a distant location, making an in-person meeting impossible. He offered to pay ₹20,000 — ₹2,000 above the asking price — if Vikram could ship the laptop. Vikram agreed.

The buyer then sent a screenshot of a Google Pay transaction, claiming he had transferred ₹20,000 to Vikram's UPI ID. He said Vikram needed to 'scan a QR code to accept the payment' — a step he claimed was required by the bank for high-value transfers to new recipients.

Vikram, eager to close the deal, scanned the QR code using his PhonePe app. Instead of receiving money, he was prompted to enter his UPI PIN to 'authorise the receipt'. He entered his PIN. Within seconds, ₹18,500 was debited from his account. The buyer's number was immediately switched off.

The fraud worked because Vikram did not understand a fundamental principle of UPI: you never need to scan a QR code or enter your UPI PIN to receive money. QR codes and PINs are used to send money — never to receive it. The 'payment received' screenshot was fabricated, and the QR code was a payment request, not a receipt confirmation.

Vikram reported the fraud to the cybercrime helpline (1930) and filed a complaint on cybercrime.gov.in. The bank was able to place a hold on ₹12,000 of the transferred amount, but ₹6,500 was irrecoverably lost.

6.2.3 How This Fraud Works: Technical Breakdown

The fraud in the case study belongs to a category known as 'QR Code Payment Request Fraud' or 'Receive Money Fraud'. Here is a technical breakdown of how it works:

36. The scammer identifies a target selling something online (OLX, Quikr, Facebook Marketplace).
37. The scammer contacts the seller, creates urgency (claiming to be in the army, abroad, or otherwise unable to meet in person), and offers a slightly higher price to build trust.
38. The scammer claims to initiate a payment and sends a fabricated screenshot as 'proof' that the money has been transferred but is 'pending confirmation'.
39. The scammer asks the seller to scan a QR code to 'accept' or 'verify' the payment. The QR code is actually a payment request generated by the scammer's UPI app.
40. When the victim scans the QR code and enters their UPI PIN, they are authorising a payment from their account to the scammer's account — the exact opposite of what they believe they are doing.
41. The money is transferred instantly. The scammer disconnects immediately.

6.2.4 Types of UPI Fraud in India

Beyond the QR code fraud illustrated in the case study, UPI fraud in India takes several other forms:

- Vishing (Voice Phishing): Fraudsters call victims posing as bank customer care representatives, IT department officials, or telecom company staff. They trick victims into revealing UPI PINs, OTPs, or bank account details by creating urgency — 'Your account will be blocked unless you verify your details right now.'
- SIM Swap Fraud: Fraudsters convince a mobile network operator to transfer a victim's mobile number to a SIM card in their possession. Once they control the number, they can receive OTPs and reset UPI accounts linked to that number.
- Phishing Links: Fraudsters send SMS or WhatsApp messages with links to fake UPI portals or banking websites. When the victim enters their credentials, the fraudster captures them and uses them to initiate fraudulent transactions.
- Fake Customer Care: Fraudsters create fake bank customer care numbers that appear at the top of Google search results (through search engine manipulation). Victims who search 'HDFC Bank customer care number' may find the fraudster's number and inadvertently share sensitive account information.
- Lottery and Prize Fraud: Victims receive messages claiming they have won a prize, lottery, or government benefit. To 'claim' the prize, they are asked to pay a

'processing fee' or 'tax' via UPI. Once paid, the fraudsters disappear and no prize is delivered.

6.2.5 Consequences of UPI Fraud

The consequences of UPI fraud operate at multiple levels:

- Financial loss: Money transferred via UPI is processed instantaneously. Unlike credit card fraud, where transactions can be disputed and reversed more easily, UPI transactions are difficult to reverse once processed.
- Psychological impact: Victims of financial fraud often experience anxiety, shame, and self-blame — particularly when the fraud occurred due to what appears, in retrospect, to be a straightforward mistake. For college students with limited financial resources, even a ₹5,000–₹20,000 loss can have significant consequences.
- Account security risks: If a fraudster obtains a victim's UPI PIN, bank account number, and mobile number, they may attempt to access other financial services linked to those credentials.
- Data privacy implications: In some fraud scenarios, victims inadvertently share personal data (Aadhaar numbers, PAN numbers, bank account details) that can be sold on the dark web or used for identity theft.

6.3 Part B: Stay Safe Online Prevention Checklist

Stay Safe Online – Prevention Checklist for College Students in India

General Online Safety:

42. Never click on links in unsolicited SMS, WhatsApp, or email messages — even if they appear to come from your bank, the government, or a familiar company. Always navigate directly to the official website.
43. Use a strong, unique password for every online account. Consider using a free password manager like Bitwarden to generate and store passwords securely.
44. Enable two-factor authentication (2FA) on all accounts that support it — especially email, social media, and banking apps.
45. Regularly check your bank and UPI transaction history for unauthorised transactions. Report any suspicious activity to your bank and the cybercrime helpline immediately.
46. Keep your mobile phone, laptop, and tablet updated with the latest software and security patches. Many cyberattacks exploit known vulnerabilities in outdated software.
47. Use only trusted, official Wi-Fi networks. Avoid conducting banking or sensitive transactions over public Wi-Fi in cafeterias, malls, or railway stations.

UPI and Financial Safety (Mandatory):

48. Remember the golden rule of UPI: You NEVER need to scan a QR code or enter your UPI PIN to receive money. If anyone tells you otherwise, it is a fraud — disconnect immediately.
49. Never share your UPI PIN, bank OTP, CVV, or full card number with anyone — not even someone claiming to be from your bank, the government, or a telecom company. Legitimate institutions will never ask for this information.
50. Before scanning any QR code for a payment, verify the amount shown on your UPI app screen. Never enter your PIN if the amount displayed is different from what you expect to pay.
51. If you are selling something online, conduct all transactions in person and request cash payment or confirm UPI receipt only after the actual amount appears in your bank account — not based on a screenshot.

Social Media and Identity Safety:

52. Review your social media privacy settings at least once per semester. Ensure personal posts are visible only to trusted friends, not to the public.
53. Do not post travel plans, hostel addresses, or location-tagged photos in real time — these can be exploited by bad actors.
54. If you witness or experience cyberbullying, do not retaliate. Document the incident (screenshots) and report it to your institution's student welfare office and, if necessary, to the cybercrime portal.

Reporting Channels:

National Cyber Crime Reporting Portal: cybercrime.gov.in

Cyber Crime Helpline: 1930 (available 24 hours a day, 7 days a week)

Local Police: Visit the nearest police station to file an FIR for financial cybercrime.

6.4 Official Reflection Write-Up (100–150 Words)

What surprised me most while researching UPI fraud was the sophistication of the social engineering involved. The fraudsters in QR code scams are not exploiting a technical vulnerability — they are exploiting a gap in the victim's understanding of how UPI works. The fraudsters know exactly what they are doing; it is the victim who is operating on false assumptions. This was a humbling realisation: being digitally active does not automatically make one digitally safe.

The one habit I will personally change is to always verify UPI transactions through my bank's official app before considering any transaction confirmed — and to never act on urgency created by an unknown caller or message. Fraudsters consistently exploit time pressure; pausing for thirty seconds to verify can prevent significant financial and emotional harm.

6.5 Extended Analysis: Cybercrime Landscape in India

6.5.1 Scale and Growth of Cybercrime in India

India's cybercrime landscape has grown dramatically in recent years, driven by the rapid expansion of internet access, smartphone adoption, and digital payment infrastructure. According to the Indian Cyber Crime Coordination Centre (I4C) and the National Crime Records Bureau (NCRB), cybercrime cases in India have increased by over 300% between 2019 and 2024.

Key statistics that contextualise the scale of the problem:

- India ranks among the top five countries globally for cybercrime volume.
- Financial fraud accounts for the majority of reported cybercrime cases in India.
- WhatsApp is reported as the most common platform through which cybercrime-related communication is initiated.
- The 1930 cybercrime helpline receives tens of thousands of calls per month, with UPI fraud and online financial scams being the most frequently reported categories.

6.5.2 The Cybercrime Ecosystem

Cybercrime in India is not random — it operates within a structured ecosystem with defined roles and infrastructure:

- Script writers: Individuals who develop the social engineering scripts used by callers to manipulate victims.
- Call centre operators: Organised call centres — some operating from foreign countries — that run large-scale fraud operations, with agents trained to impersonate bank officials, government representatives, or telecom staff.
- Money mules: Individuals (sometimes unwitting) who allow their bank accounts to be used to receive and launder fraud proceeds.
- Technical operators: Individuals who set up fake websites, create phishing links, and manage the technical infrastructure of fraud campaigns.
- Dark web marketplaces: Platforms where stolen financial credentials, personal data, and fraud tools are bought and sold.

6.5.3 Cyberbullying: A Growing Concern Among Students

While UPI fraud was the focus of this case study, cyberbullying deserves special attention as a form of cybercrime particularly prevalent among college students.

Cyberbullying is defined as the use of digital technology — social media, messaging apps, online forums — to harass, intimidate, threaten, or humiliate an individual. Unlike traditional bullying, cyberbullying:

- Has no geographic boundaries — it can reach victims at home, at night, and in spaces where they previously felt safe.
- Can be anonymous — perpetrators can hide behind fake accounts or anonymous platforms.
- Is permanent and replicable — screenshots and recordings of bullying content can be shared widely and are difficult to remove completely.
- Has a wide audience — humiliating content can be seen by hundreds or thousands of people instantly.

The psychological impact of cyberbullying is well-documented and includes anxiety, depression, social withdrawal, academic underperformance, and, in severe cases, self-harm and suicidal ideation. Students who experience cyberbullying should:

- Document the bullying with screenshots before blocking or reporting the perpetrator.
- Report the content to the platform (Instagram, Twitter/X, WhatsApp).
- Report the incident to the university's student welfare office and, if the bullying constitutes criminal harassment, to the police or cybercrime portal.
- Seek support from trusted friends, family, or a counsellor.

6.5.4 Ransomware: A Growing Threat to Institutions

Ransomware is a form of malicious software (malware) that encrypts the victim's files and demands a ransom payment — typically in cryptocurrency — in exchange for the decryption key. While ransomware primarily targets organisations (hospitals, banks, government agencies, educational institutions), individual students can also be targeted, particularly if they download software from unverified sources or open malicious email attachments.

For college students, the most relevant ransomware prevention measures are:

- Never download software from unofficial websites or torrent sites. Use official, verified sources for all software installations.
- Back up important files regularly — to an external hard drive and/or a cloud service. If ransomware encrypts your files, a recent backup allows recovery without paying the ransom.
- Do not open email attachments from unknown senders, particularly .exe, .zip, .doc (with macros enabled), or .pdf files from unfamiliar sources.
- Keep your operating system and antivirus software updated.

6.5.5 Identity Theft: What It Means and Why It Matters

Identity theft occurs when someone fraudulently obtains and uses another person's personal information — typically for financial gain. In the digital context, identity theft can involve:

- Using a stolen Aadhaar number and PAN card to open fraudulent bank accounts or apply for loans.
- Creating fake social media profiles using someone's name and photos to deceive others.
- Using stolen credentials to access email, cloud storage, or online banking accounts.

For college students, identity theft risks arise from:

- Oversharing personal information on social media (full name, date of birth, phone number, college name).
- Using the same email and password combination across multiple platforms — a single data breach can compromise all linked accounts.
- Filling in online forms (for 'free gifts', contest entries, or survey participation) that collect personal data without clear privacy policies.

7. GitHub Repository Structure and Documentation

7.1 Repository Overview

The complete Digital Literacy Portfolio project is organised in a single public GitHub repository named digital-literacy-project. The repository serves as both the delivery mechanism for all task deliverables and as a demonstration of basic version control and digital organisation skills.

Repository URL: https://github.com/rahul-vit-bce/digital-literacy-project
Visibility: Public
Primary Branch: main
Created: 15 January 2026
Last Updated: 29 March 2026

7.2 Folder Structure

The repository is organised according to the structure specified in the project brief:

- README.md – Project overview, student details, module summaries, links to task folders, and the Google Form quiz link.
- report/ – Contains this project report (Project_Report.docx and Project_Report.pdf).
- task-1-presentation/ – Contains the Canva infographic (digital_literacy_infographic.png) and a description file (description.md).
- task-2-portfolio/ – Contains screenshots of the GitHub profile README, LinkedIn Education section, and Kaggle profile page.
- task-3-platforms/ – Contains screenshots of the completed HackerRank challenge, the Google Form in preview mode, and the linked Google Sheet response view.
- task-4-email-etiquette/ – Contains email_drafts.md (both email drafts) and social-media-checklist.md.
- task-5-cybercrime/ – Contains casestudy.md (the UPI fraud case study) and prevention-checklist.md.

7.3 README.md Content

The README.md file at the root of the repository contains the following sections:

55. Project Title and Introduction: A brief overview of the project scenario (Student Digital Ambassador).

56. Student Details: Name, registration number, branch, year, course code, and date of submission.
57. Repository Structure: A table listing each folder and its contents.
58. Module Summaries: A brief one-paragraph summary of each of the five tasks.
59. Task Links: Navigation links to each task folder.
60. Google Form Link: The shareable link to the Digital Literacy Awareness Quiz (Task 3, Part B).
61. References: Links to key platforms and tools used throughout the project.

8. Conclusion

8.1 Overall Learning Outcomes

The Digital Literacy Portfolio project has been one of the most practically valuable assignments of my first semester at VIT Bhopal University. Unlike most coursework, which tests the ability to reproduce knowledge in an examination setting, this project required active engagement with real digital tools, real platforms, and real-world challenges.

Across the five tasks, I developed competencies in five distinct dimensions of digital literacy:

- Visual communication (Task 1): Creating an infographic taught me that clarity and simplicity in design are skills that require deliberate practice — not just aesthetic instinct.
- Digital identity management (Task 2): Setting up GitHub, LinkedIn, and Kaggle gave me a tangible sense of what a professional digital presence looks like and what it takes to maintain it.
- Technical platform fluency (Task 3): Using HackerRank and Google Forms introduced me to two categories of tools — coding practice platforms and cloud collaboration tools — that I will use extensively throughout my degree.
- Professional communication (Task 4): Drafting professional emails and the social media checklist made me more conscious of how tone, structure, and word choice affect professional credibility in digital communication.
- Cybersecurity awareness (Task 5): Researching UPI fraud gave me a sobering understanding of the sophisticated social engineering techniques used by fraudsters and the importance of cultivating scepticism and verification habits.

8.2 Skills Acquired

The following specific skills were acquired or strengthened through this project:

62. Canva: Designing infographics using templates, icons, and layout tools.
63. GitHub: Creating and managing repositories, editing README files, uploading files via the browser.
64. LinkedIn: Setting up and maintaining a professional profile with accurate education information.
65. Kaggle: Navigating the platform and understanding its learning and competition resources.
66. HackerRank: Completing beginner-level programming challenges in Python.
67. Google Forms: Creating a multi-question quiz with multiple-choice and short-answer questions.

68. Google Sheets: Understanding how Google Forms responses are automatically collected in a linked spreadsheet.
69. Professional email writing: Applying subject line, greeting, body, and sign-off conventions to two formal emails.
70. Content planning: Organising ideas before creating any deliverable — whether an infographic, an email, or a repository structure.
71. Research: Gathering information about cybercrime scenarios, digital literacy concepts, and platform capabilities from reliable sources.

8.3 Future Plans

The habits and skills developed through this project are intended to compound over time. Here is how I plan to build upon each task going forward:

Building on Task 1 (Visual Communication):

I plan to use Canva and similar tools to create visual summaries of complex technical concepts as I progress through my degree. Being able to visualise a data structure, an algorithm, or a system architecture diagram clearly is a valuable skill for technical communication and project presentations.

Building on Task 2 (Digital Portfolio):

I will update my GitHub, LinkedIn, and Kaggle profiles at the end of each semester with new projects, certifications, and achievements. By the time I am a third-year student applying for internships, these profiles should tell a coherent story of consistent learning and growth.

Building on Task 3 (Platforms):

I will continue practising on HackerRank, progressively working through intermediate and advanced problem categories. I plan to earn the Python Basic certification by the end of Year 1 and the Problem Solving (Basic) certification by the end of Year 2.

Building on Task 4 (Communication):

The email writing skills developed in this task will be applied immediately — in my first-year faculty communications, club applications, and, eventually, internship applications. The social media checklist will be shared with my batch through the Student Digital Ambassador programme.

Building on Task 5 (Cybersecurity):

I will share the prevention checklist with family members — particularly those who are new to UPI and digital payments. I will also explore additional cybersecurity

resources, including the Indian Computer Emergency Response Team (CERT-In) advisories and the NPTEL course on Ethical Hacking as my skills develop.

In summary, this project has not merely produced a set of deliverables — it has established habits, practices, and a mindset that I intend to carry forward throughout my degree and professional career. Being digitally literate in 2026 means being continuously curious, continuously learning, and continuously questioning. This portfolio is the beginning of that journey.

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[19] Information Technology Act, 2000. Indian law governing cybercrime and digital communication, referenced in Task 4 extended analysis. Available at: https://www.meity.gov.in

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Appendix A: Glossary of Digital Literacy Terms

Two-Factor Authentication (2FA): A security mechanism that requires two forms of identification before granting access to an account — typically a password plus a one-time code sent via SMS or generated by an authenticator app.

Cyberbullying: The use of digital technology — social media, messaging apps, online forums — to harass, intimidate, threaten, or humiliate an individual.

Cybercrime: Criminal activity that involves a computer, networked device, or a network as either the target of the crime or the tool used to commit it.

Dark Web: A part of the internet that is not indexed by standard search engines and requires special software (such as the Tor browser) to access. It is used for both legitimate privacy-protecting purposes and illegal activities.

Data Literacy: The ability to read, interpret, create, and communicate data meaningfully.

Digital Footprint: The trail of data created while using the internet. It includes websites visited, emails sent, social media activity, and online purchases. A digital footprint can be active (deliberately shared information) or passive (data collected without the user's knowledge).

Digital Identity: The online representation of an individual, composed of profiles, content, and interactions across digital platforms.

Digital Literacy: The ability to access, manage, understand, integrate, communicate, evaluate, and create information safely and appropriately through digital technologies.

FIR (First Information Report): A written document prepared by the police when they receive information about a cognisable offence. Cybercrime victims can file an FIR at their local police station.

Git: A distributed version control system that tracks changes to files over time, allowing multiple users to collaborate on the same project.

GitHub: A web-based platform built on Git that provides hosting for software development projects, version control, and collaboration features.

Infographic: A visual representation of information or data, designed to present complex information quickly and clearly.

Information Technology Act, 2000: The primary Indian law governing cybercrime, electronic commerce, and digital communication.

Malware: Malicious software designed to disrupt, damage, or gain unauthorised access to a computer system. Includes viruses, worms, trojans, ransomware, and spyware.

OTP (One-Time Password): A password valid for only one login session or transaction, typically sent via SMS. OTPs are used as a second factor in two-factor authentication.

Password Manager: A software application that securely stores and organises passwords, generates strong random passwords, and auto-fills them on websites and apps.

Phishing: A fraudulent attempt to obtain sensitive information (passwords, bank details, OTPs) by disguising as a trustworthy entity in electronic communication.

Profile README: A special GitHub repository named after a user's GitHub username, the content of which is displayed on their public GitHub profile page.

QR Code: A machine-readable matrix barcode that stores information, typically a URL or payment request. In UPI, QR codes can be used to initiate payments.

Ransomware: A type of malware that encrypts the victim's files and demands a ransom payment in exchange for the decryption key.

SIM Swap Fraud: A fraud in which an attacker convinces a mobile network operator to transfer a victim's phone number to a new SIM card, allowing the attacker to receive OTPs and reset accounts.

Smishing: Phishing conducted via SMS messages.

Social Engineering: Psychological manipulation of individuals into performing actions or divulging confidential information. Most cybercrime relies on social engineering rather than purely technical exploits.

UPI (Unified Payments Interface): A real-time payment system developed by the National Payments Corporation of India (NPCI) that facilitates instant fund transfers between bank accounts using mobile applications.

Version Control: A system that records changes to files over time, allowing specific versions to be recalled later. Git is the most widely used version control system.

Vishing: Phishing conducted via voice calls (phone calls).

Appendix B: Useful Resources and Links

Task 1 – Infographic Design Tools

Canva (canva.com): Browser-based graphic design tool, recommended for infographics and presentations.

Piktochart (piktochart.com): Specialised infographic creation tool with data visualisation features.

Prezi (prezi.com): Non-linear presentation and visual storytelling tool.

SlidesCarniVal (slidescarnival.com): Free PowerPoint and Google Slides templates.

Visme (visme.co): Professional infographic and data visualisation tool.

Task 2 – Digital Portfolio Platforms

GitHub (github.com): Code hosting, version control, and developer portfolio platform.

LinkedIn (linkedin.com): Professional networking and career development platform.

Kaggle (kaggle.com): Data science community with datasets, competitions, and free courses.

Stack Overflow (stackoverflow.com): Developer Q&A community and knowledge base.

ResearchGate (researchgate.net): Platform for sharing and discovering academic research.

Task 3 – Coding and Collaboration Platforms

HackerRank (hackerrank.com): Coding challenges, skill certifications, and hiring platform.

CodeChef (codechef.com): Competitive programming contests and practice problems.

LeetCode (leetcode.com): Technical interview preparation with algorithm problems.

Codeforces (codeforces.com): Competitive programming platform with rated contests.

Google Forms (forms.google.com): Free online form and survey builder.

Google Sheets (sheets.google.com): Cloud-based spreadsheet application.

Google Drive (drive.google.com): Cloud storage for Google Workspace files.

Task 4 – Communication Resources

Grammarly (grammarly.com): Writing assistance tool for grammar, spelling, and tone checking.

Hemingway Editor (hemingwayapp.com): Tool for improving writing clarity and readability.

Purdue OWL (owl.purdue.edu): Online Writing Lab with guides on professional and academic writing.

Task 5 – Cybersecurity Resources

National Cyber Crime Reporting Portal (cybercrime.gov.in): Government of India portal for reporting cybercrime.

CERT-In (cert-in.org.in): Indian Computer Emergency Response Team – security advisories and alerts.

RBI Kehta Hai (rbi.org.in/kehta-hai): Reserve Bank of India consumer awareness resources for financial safety.

Stay Safe Online (NCSA) (staysafeonline.org): Global cybersecurity awareness resources.

Cyber Dost (MHA) (twitter.com/CyberDost): Ministry of Home Affairs cybercrime awareness initiative.

General Learning Platforms

NPTEL (nptel.ac.in): National Programme on Technology Enhanced Learning – free Indian university courses.

Coursera (coursera.org): Online courses from top global universities and companies.

edX (edx.org): Online learning platform with courses from MIT, Harvard, and others.

Khan Academy (khanacademy.org): Free educational resources across subjects including computing and programming.

freeCodeCamp (freecodecamp.org): Free programming curriculum covering web development and data science.

Appendix C: Self-Assessment Rubric

The following self-assessment rubric reflects my honest evaluation of my own performance across each task and component of the project.

Task 1 – Infographic: [Self-assessed: 18/20]

Infographic created using Canva. Three topics covered (digital literacy definition, safe internet practices, professional online presence). Design is clean and readable. Reflection write-up is within the 150–200 word range and addresses all required points.

Task 2 – Digital Portfolio: [Self-assessed: 19/20]

Accounts created on GitHub (profile README + project repo), LinkedIn (Education section complete), and Kaggle (profile set up, courses explored). Screenshots taken and uploaded. Reflection write-up within word limit and covers all required points.

Task 3 – Platforms: [Self-assessed: 19/20]

HackerRank: Two beginner challenges completed (Solve Me First + Python Division). Both accepted. Screenshots captured. Google Form: Five questions created (3 MCQ + 2 short answer). Form linked to Google Sheet. Screenshots captured. Reflection write-up within word limit.

Task 4 – Email Etiquette: [Self-assessed: 18/20]

Two professional emails drafted with correct subject lines, greetings, body structure, and sign-offs. Social media checklist created with 7 Do's and 7 Don'ts. Communication scenario reflection within 100–150 words.

Task 5 – Cybercrime: [Self-assessed: 19/20]

Case study on UPI fraud written with realistic fictional scenario. Covers what the crime is, how it happens, who is targeted, and consequences. Word count within 200–250. Prevention checklist has 13 tips (exceeds minimum of 8). Two UPI-specific tips included. Reporting channels cited. Reflection within 100–150 words.

Repository Structure: [Self-assessed: Bonus: 5/5]

All required folders present. README.md complete with overview, student details, module summaries, task links, and Google Form link. Repository is public.

Self-Assessed Total: 93/100 + 5 Bonus = 98/100

End of Report